Mehdi Sadeghi

Email | LinkedIn | GitHub

Education

Arizona State University — Temep, AZ
Ph.D., Civil Engineering — 2026 - present
- Research area: Numerical modeling and data driven solutions

Oklahoma State University — Stillwater, OK
Ph.D., Civil Engineering — 2023 - 2025
- Research area: Asphalt materials testing and data-driven prediction modeling

Sharif University of Technology — Tehran, Iran
M.S., Civil Engineering — 2018 - 2022
- Research area: Pavement design and life cycle analysis

Azad University — Isfahan, Iran
B.S., Civil Engineering — 2013 - 2018

Research Interests

Asphalt materials characterization, programming, software development, pavement modeling, and sustainability

Academic Experience

Research Associate | Arizona State University — 2026 - present
- Data Analysis and Machine Learning, Programming, Image Processing and Computer Vision, Tool Development, Asphalt Materials Characterization

Research Assistant | Oklahoma State University — 2023 - 2025
- Asphalt Materials Characterization (Binder, RAP, WMA, PFC), Data Analysis and Machine Learning, Programming, Tool Development

Teaching Assistant | Oklahoma State University — 2023 - present
- Engineering Materials Lab

Research Assistant | Sharif University of Technology — 2019 - 2022
- MEPDG Pavement Design, Rehabilitation Techniques, Sustainability, LCCA, LCA, Programming, Tool Development

Teaching Assistant | Sharif University of Technology — 2019 - 2023
- Advanced Highway Pavement Design, Asphalt Laboratory, Pavement Management Systems, and Concrete Pavement Design

Skills

Programming: Python, R, LaTeX, VBA
Software: AASHTOWare PMED, OpenLCA, ABAQUS, JMP Pro, and MS Office
Other: Research publication, proposal writing, and lab setup and management
Languages: English and Persian

Technical Reports

Incorporating recycled reclaimed asphalt pavement into the balanced mix design world
Oklahoma Department of Transportation 2023 - 2025

Performance-based design of permeable friction courses using warm mix asphalt for enhanced safety and durability
Southern Plains Transportation Center 2024 - 2025

Investigate the aging behavior of asphalt binders at different production stages and during the service life of the pavement
Oklahoma Department of Transportation 2022 - 2025

Cold reclamation and recycling techniques to achieve perpetual pavements: Task 1
Minnesota Department of Transportation 2022 - 2025

Journal Publications

1. S. Mousavi Rad, **M. Sadeghi**, J. Bausano, D. Vivanco, & M. Elkashef. Evaluation of asphalt mixture cracking resistance and development of a machine learning-based application for cracking tolerance index prediction. *Construction and Building Materials, 490:142519*, 2025. DOI.
2. M. Olagunju, S. Mousavi Rad, **M. Sadeghi**, D. Vivanco, & M. Elkashef. Evaluating the rutting and moisture resistance of asphalt mixes in Oklahoma using indirect tensile test at high temperature strength and Hamburg wheel tracking test corrected rut depth and stripping number. *Transportation Research Record: 03611981251324206*, 2025. DOI.
3. P. Akinyoola, H. Hassan, **M. Sadeghi**, D. Vivanco, & M. Elkashef. Selection of a laboratory short-term aging protocol to evaluate cracking resistance of asphalt mixes considering production variability. *Construction and Building Materials*. [Accepted for Publication]
4. **M. Sadeghi**, R. Steger, & M. Elkashef. Evaluating fibreless warm mix asphalt permeable friction courses using performance-related testing. *International Journal of Pavement Engineering*. [Under Review]
5. **M. Sadeghi**, S. Mousavi Rad, & M. Elkashef. Analyzing the cracking behavior of asphalt mixtures using principal component analysis and k-means clustering. *Journal of Testing and Evaluation*. [Under Review]
6. **M. Sadeghi** & M. Sabouri. Multi-Criteria Comparative Sustainability Assessment of Perpetual Pavements in Iran Using MEPDG Design Approach. *Scientia Iranica*. [Under Review]

Conference Presentation

1. **M. Sadeghi** & M. Elkashef. Cracking resistance estimating tool. (Demo Presentation). In 2025 OTRD. Edmond, OK.
2. S. Mousavi Rad, **M. Sadeghi**, J. Bausano, D. Vivanco, & M. Elkashef. Evaluation of asphalt mixture cracking resistance and development of a machine learning-based application for cracking tolerance index prediction. (Poster Session). In 2025 OTRD. Edmond, OK.
3. **M. Sadeghi**, R. Steger, & M. Elkashef. Enhancing porous friction course design: Incorporating warm Mix asphalt and performance-based design. (Podium Session). In 2025 International Airfield and Highway Pavements Conference. Glendale, AZ.
4. **M. Sadeghi**, S. Mousavi Rad, & M. Elkashef. Using principal component analysis and k-means clustering to analyze the cracking behavior of asphalt mixtures. (Poster Session). In 2025 Annual TRB Conference. Washington, DC.
5. M. Olagunju, S. Mousavi Rad, **M. Sadeghi**, D. Vivanco, & M. Elkashef. Evaluating the rutting and moisture resistance of asphalt mixes in Oklahoma using IDT-HT strength and HWTT corrected rut depth and stripping number. (Podium Session). In 2025 Annual TRB Conference. Washington, DC.
6. H. Hassan, **M. Sadeghi**, & M. Elkashef. A case study on production variability of BMD and Superpave mixes with RAP. (Poster Session). In 2024 OTRD. Edmond, OK.
7. M. Olagunju, S. Mousavi Rad, **M. Sadeghi**, & M. Elkashef. Study of rutting and moisture resistance of asphalt mixes in Oklahoma using HWTT and IDT-HT. (Poster Session). In 2024 OTRD. Edmond, OK.
8. **M. Sadeghi**, S. Mousavi Rad, & M. Elkashef. Using principal component analysis and k-means clustering to analyze the cracking behavior of asphalt mixtures. (Poster Session). In 2024 OTRD. Edmond, OK.

Programming Projects

Pavement Sustainability Assessment Tool
A Python-based desktop application designed to aid in selecting sustainable pavement construction alternatives, integrating LCCA, LCA, and performance evaluation. SUT, 2022

Asphalt Cracking and Rutting Performance Estimation Tool
A web-based platform and database management system that uses machine learning to predict pavement performance based on volumetric properties of asphalt mixes. Developed using the ODOT and OSU mix testing database, the tool estimates cracking and rutting potential from mix design parameters. OSU, 2025

Certifications

- **GRE General**, Verbal: 152, Quantitative: 161, Analytical Writing: 3.5 — 2021
- **Mastering Bitumen for Better Roads and Innovative Applications** — Coursera, 2023
- **Construction Life Cycle Assessment Specialist** — OneClickLCA, 2023
- **Neural Networks and Deep Learning** — Coursera, 2023
- **Supervised Machine Learning: Regression and Classification** — Coursera, 2023

Academic Services

Reviewer: Transportation Research Record, Construction and Building Materials, International Journal of Pavement Engineering, International Journal of Pavement Research and Technology

Member: American Society of Civil Engineers (ASCE), Tau Beta Pi Engineering Honor Society